	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\ \dagger}$	0	0
$\mathcal{K}_{0^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{K}_{1^+}^{\#1\ \alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\ \alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\ \alpha}$	$\mathcal{K}_{1^-}^{\#2\ \alpha}$
$\mathcal{K}_{1^+}^{\#1\ \dagger\alpha\beta}$	$-k^2\ {}^{(4)}\kappa_3$	$\frac{k^2\ {}^{(4)}\kappa_4}{2\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2\ \dagger\alpha\beta}$	$\frac{k^2\ {}^{(4)}\kappa_4}{2\sqrt{2}}$	$\frac{\Upsilon_1 k^2}{2}$	0	0
$\mathcal{K}_{1^-}^{\#1\ \dagger\alpha}$	0	0	$k^2\ {}^{(4)}\kappa_1$	$\sqrt{2}\ k^2\ {}^{(4)}\kappa_1$
$\mathcal{K}_{1^-}^{\#2\ \dagger\alpha}$	0	0	$\sqrt{2}\ k^2\ {}^{(4)}\kappa_1$	$2\ k^2\ {}^{(4)}\kappa_1$

	$\mathcal{K}_{2^+}^{\#1\ \alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1\ \dagger\alpha\beta}$	0	0
$\mathcal{K}_{2^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\ \dagger}$	0	0
$\mathcal{J}_{0^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{1^+}^{\#1\ \alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\ \alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\ \alpha}$	$\mathcal{J}_{1^-}^{\#2\ \alpha}$
$\mathcal{J}_{1^+}^{\#1\ \dagger\alpha\beta}$	$\frac{\Upsilon_1 k^2}{2\ \mathrm{Det}(1^+)}$	$-\frac{k^2\ {}^{(4)}\kappa_4}{2\sqrt{2}\ \mathrm{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2\ \dagger\alpha\beta}$	$-\frac{k^2\ {}^{(4)}\kappa_4}{2\sqrt{2}\ \mathrm{Det}(1^+)}$	$-\frac{k^2\ {}^{(4)}\kappa_3}{\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}_{1^-}^{\#1\ \dagger\alpha}$	0	0	$\frac{1}{9k^2\ {}^{(4)}\kappa_1}$	$\frac{\sqrt{2}}{9k^2\ {}^{(4)}\kappa_1}$
$\mathcal{J}_{1^-}^{\#2\ \dagger\alpha}$	0	0	$\frac{\sqrt{2}}{9k^2\ {}^{(4)}\kappa_1}$	$\frac{2}{9k^2\ {}^{(4)}\kappa_1}$

	$\mathcal{J}_{2^+}^{\#1\ \alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\ \dagger\alpha\beta}$	0	0
$\mathcal{J}_{2^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

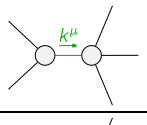
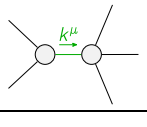
Abbreviations used in matrices

$$\Upsilon_1 == 9\ {}^{(4)}\kappa_1 + {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4\ \&\&\ \mathrm{Det}(1^+) == -\frac{1}{8}k^4\left(36\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_3 + (2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2\right)$$

Lagrangian

$$\begin{aligned}&{}^{(4)}\kappa_1\ \partial_\beta\mathcal{K}^\delta{}_{\chi\delta}\partial^\chi\mathcal{K}^{\alpha\ \beta}{}_{\alpha} - {}^{(4)}\kappa_1\ \partial_\chi\mathcal{K}^\delta{}_{\beta\delta}\partial^\chi\mathcal{K}^{\alpha\ \beta}{}_{\alpha} + {}^{(4)}\kappa_3\ \partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + {}^{(4)}\kappa_4\ \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - {}^{(4)}\kappa_3\ \partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \\ &6\ {}^{(4)}\kappa_1\ \partial^\chi\mathcal{K}^{\alpha\ \beta}{}_{\alpha}\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} + \frac{9}{2}\ {}^{(4)}\kappa_1\ \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + \frac{1}{2}\ {}^{(4)}\kappa_3\ \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + \frac{1}{2}\ {}^{(4)}\kappa_4\ \partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\ \alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha\ \beta}{}_{\alpha} == 0$
$\mathcal{J}_{1^-}^{\#1\ \alpha} == \mathcal{J}_{1^-}^{\#2\ \alpha}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta\ \chi}{}_{\beta} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}_{2^-}^{\#1\ \alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}_{2^+}^{\#1\ \alpha\beta} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	15	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{48\ {}^{(4)}\kappa_1^{\ 2} + 8\ {}^{(4)}\kappa_1(2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2}{{}^{(4)}\kappa_1(36\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_3 + (2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)}$
	1	0	$-\frac{216\ {}^{(4)}\kappa_1^{\ 2} + 24\ {}^{(4)}\kappa_1(2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2}{{}^{(4)}\kappa_1(36\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_3 + (2\ {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)}$

Resolved unitarity condition(s):

$$\begin{aligned}&({}^{(4)}\kappa_4 < 0\ \&\&\ (({}^{(4)}\kappa_3 < 0\ \&\&\ ({}^{(4)}\kappa_1 < 0 \,||\, {}^{(4)}\kappa_1 > \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3})) \,||\, ({}^{(4)}\kappa_3 == 0\ \&\&\ {}^{(4)}\kappa_1 < 0) \,||\\ &(0 < {}^{(4)}\kappa_3 < -\tfrac{{}^{(4)}\kappa_4}{2}\ \&\&\ \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3} < {}^{(4)}\kappa_1 < 0) \,||\, ({}^{(4)}\kappa_3 > -\tfrac{{}^{(4)}\kappa_4}{2}\ \&\&\ \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3} < {}^{(4)}\kappa_1 < 0))) \,||\\ &({}^{(4)}\kappa_4 == 0\ \&\&\ (({}^{(4)}\kappa_3 < 0\ \&\&\ ({}^{(4)}\kappa_1 < 0 \,||\, {}^{(4)}\kappa_1 > -\tfrac{{}^{(4)}\kappa_3}{9})) \,||\, ({}^{(4)}\kappa_3 > 0\ \&\&\ -\tfrac{{}^{(4)}\kappa_3}{9} < {}^{(4)}\kappa_1 < 0))) \,||\\ &({}^{(4)}\kappa_4 > 0\ \&\&\ (({}^{(4)}\kappa_3 < -\tfrac{{}^{(4)}\kappa_4}{2}\ \&\&\ ({}^{(4)}\kappa_1 < 0 \,||\, {}^{(4)}\kappa_1 > \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3})) \,||\\ &({}^{(4)}\kappa_3 == -\tfrac{{}^{(4)}\kappa_4}{2}\ \&\&\ ({}^{(4)}\kappa_1 < 0 \,||\, {}^{(4)}\kappa_1 > 0)) \,||\, (-\tfrac{{}^{(4)}\kappa_4}{2} < {}^{(4)}\kappa_3 < 0\ \&\&\ ({}^{(4)}\kappa_1 < 0 \,||\, {}^{(4)}\kappa_1 > \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3})) \,||\\ &({}^{(4)}\kappa_3 == 0\ \&\&\ {}^{(4)}\kappa_1 < 0) \,||\, ({}^{(4)}\kappa_3 > 0\ \&\&\ \tfrac{-4\ {}^{(4)}\kappa_3^{\ 2} - 4\ {}^{(4)}\kappa_3\ {}^{(4)}\kappa_4 - {}^{(4)}\kappa_4^{\ 2}}{36\ {}^{(4)}\kappa_3} < {}^{(4)}\kappa_1 < 0)))\end{aligned}$$